

Photosynthesis for High School Students

An introduction to the process that powers life on Earth.



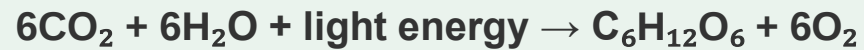
Presented by: [Your Name]

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What is Photosynthesis?

Photosynthesis is the process by which green plants, algae, and some bacteria **use sunlight to convert water and carbon dioxide** into food (glucose) and oxygen.

Photosynthesis Equation:



Carbon dioxide + Water + Sunlight → Glucose + Oxygen

Key Facts:



Occurs in chloroplasts



Requires sunlight



Produces glucose & oxygen



The Process of Photosynthesis

1. Light Reactions

Location: Thylakoid membranes

Requires: Sunlight, water, chlorophyll

Produces: ATP, NADPH, oxygen

In this stage, chlorophyll captures light energy, which is used to split water molecules (photolysis), releasing oxygen as a byproduct. The energy is stored as ATP and NADPH, which power the Calvin Cycle.



2. Dark Reactions (Calvin Cycle)

Location: Stroma of chloroplast

Requires: ATP, NADPH, CO₂

Produces: Glucose (sugar)

The Calvin Cycle doesn't directly need light, but uses the products of the light reactions. Carbon dioxide is combined with a 5-carbon compound (RuBP) to eventually form glucose. This process is also called carbon fixation.



Complete Process Flow:

Importance & Real-World Applications

Why Photosynthesis Matters



Oxygen Production

Produces approximately 70% of Earth's oxygen, making it essential for all aerobic organisms to breathe and survive.



Foundation of Food Chains

Creates the energy-rich compounds that fuel nearly all ecosystems, serving as the base of the food pyramid.



Climate Regulation

Removes CO₂ from the atmosphere, helping mitigate climate change and global warming effects.

Real-World Applications



Agriculture

Research on photosynthesis efficiency leads to crop improvements, increased yields, and food security solutions.



Renewable Energy

Biofuels derived from photosynthetic organisms offer sustainable alternatives to fossil fuels.



Scientific Research

Studies of photosynthesis inform climate change models, ecosystem health monitoring, and artificial photosynthesis technology.

"Photosynthesis doesn't just keep plants alive—it keeps our entire planet thriving."